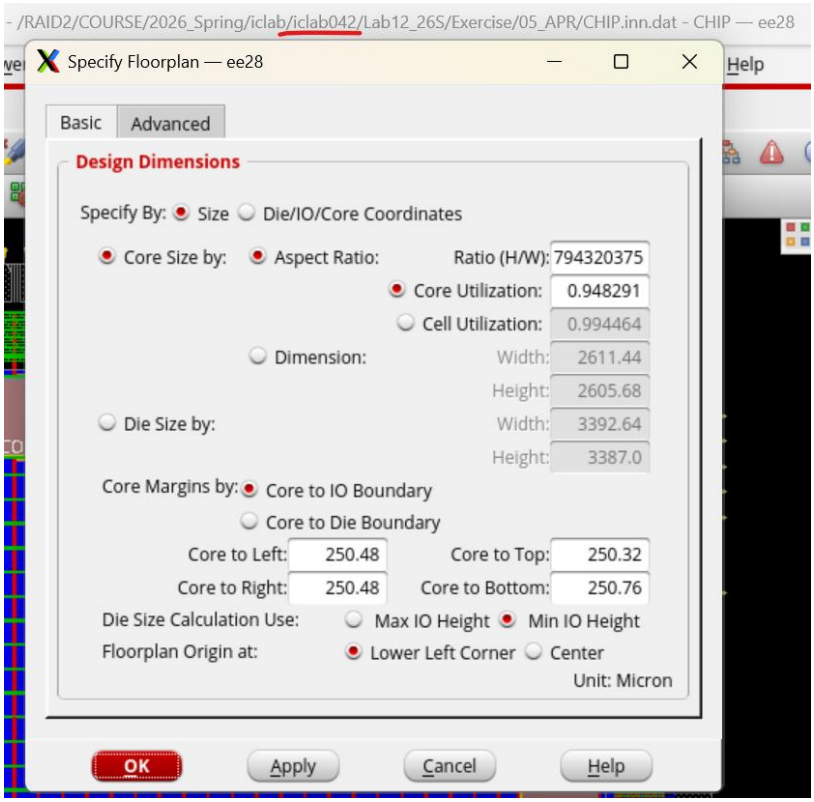
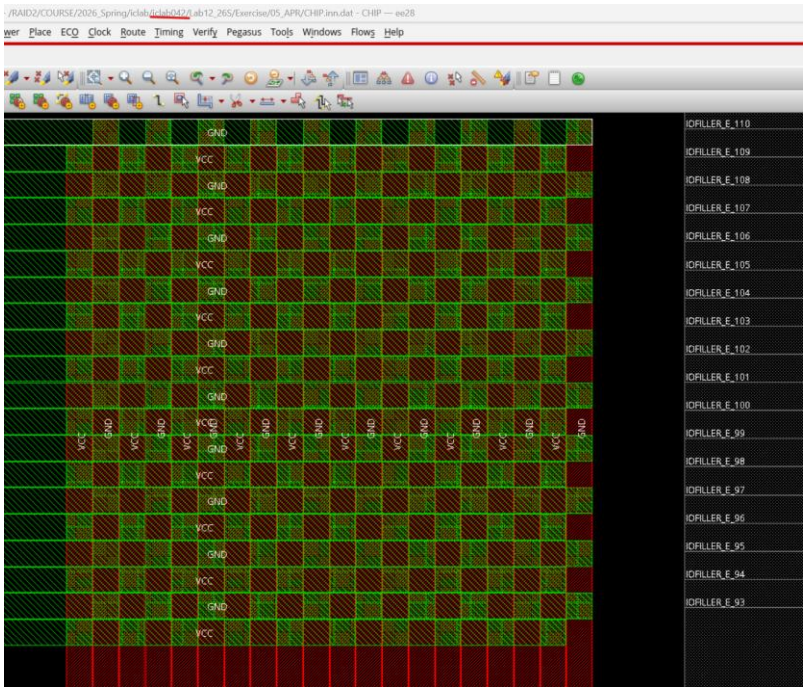
Report

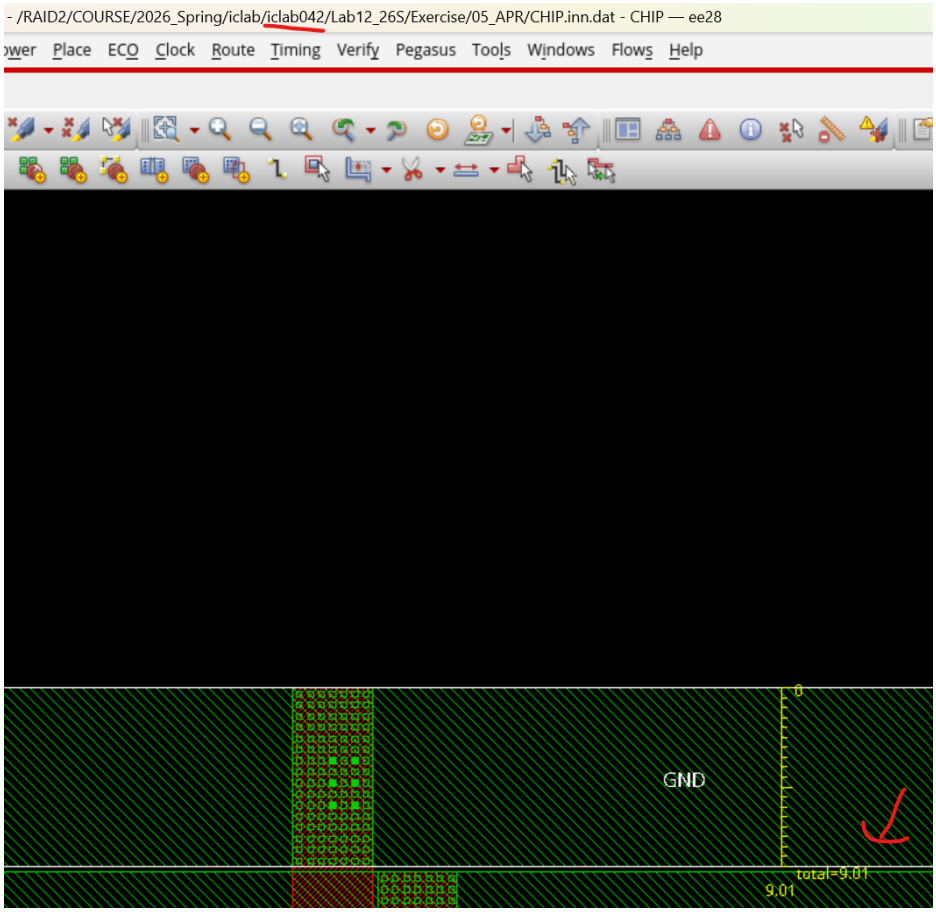
1. Core to IO boundary :



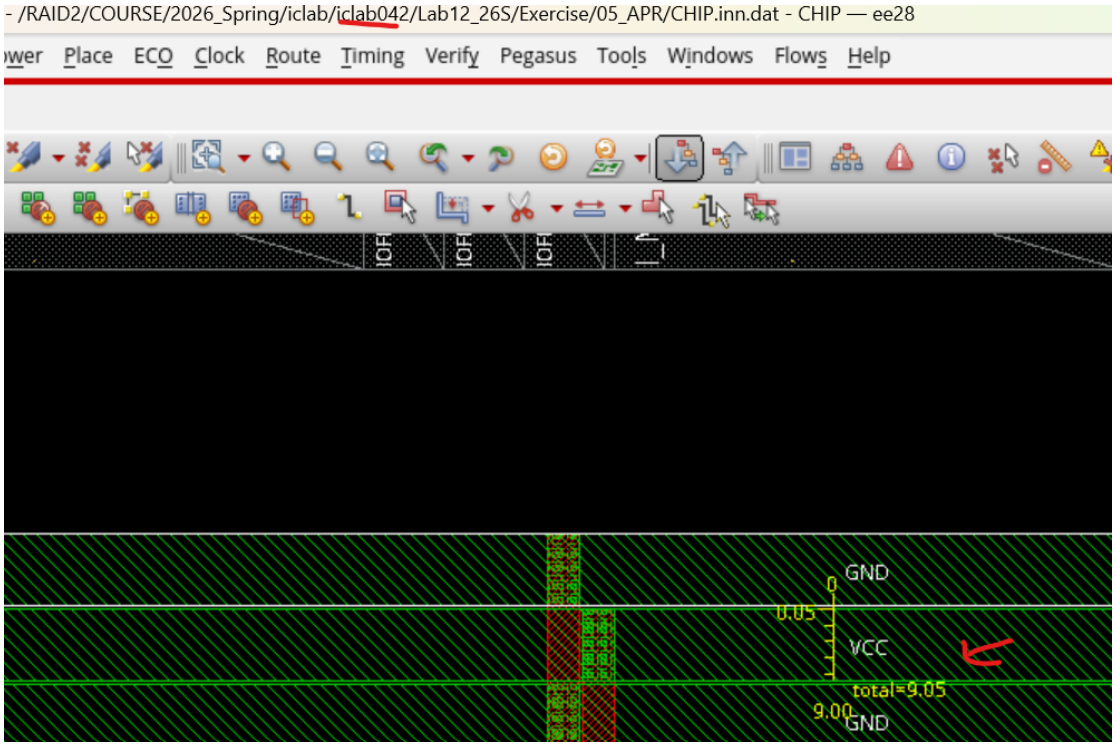
2. Core Ring :



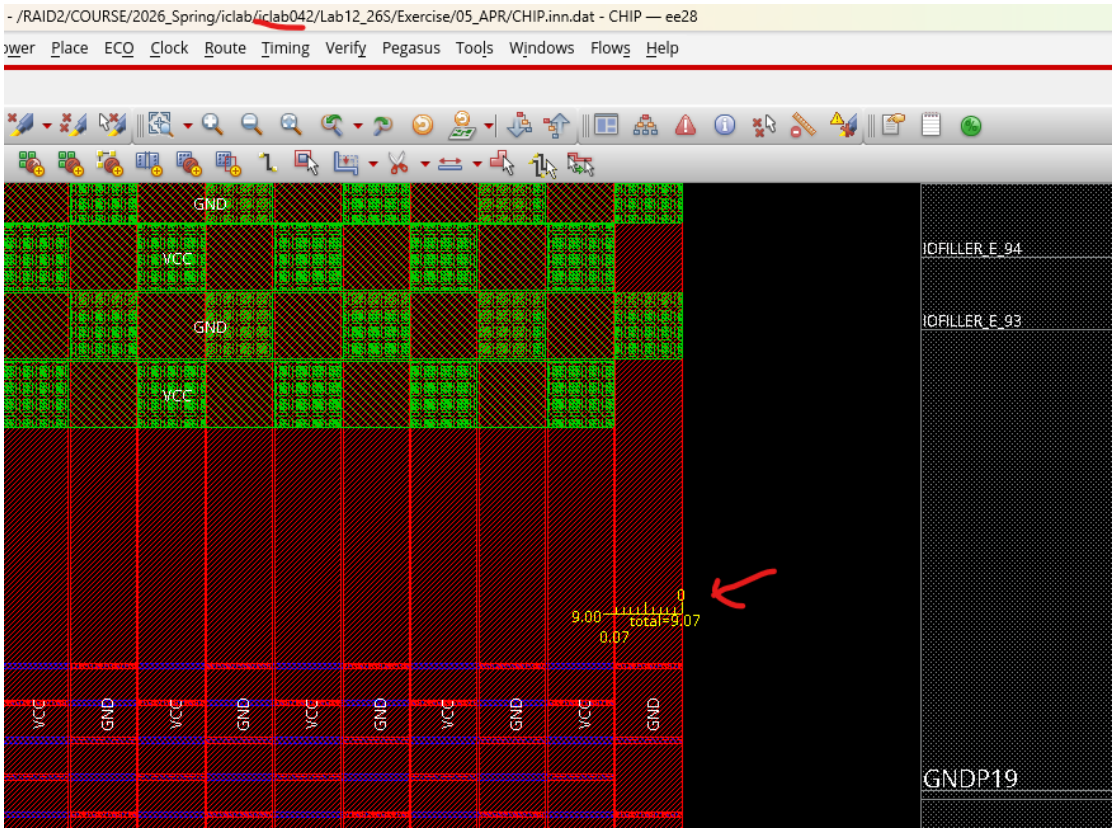
Horizontal core ring GND



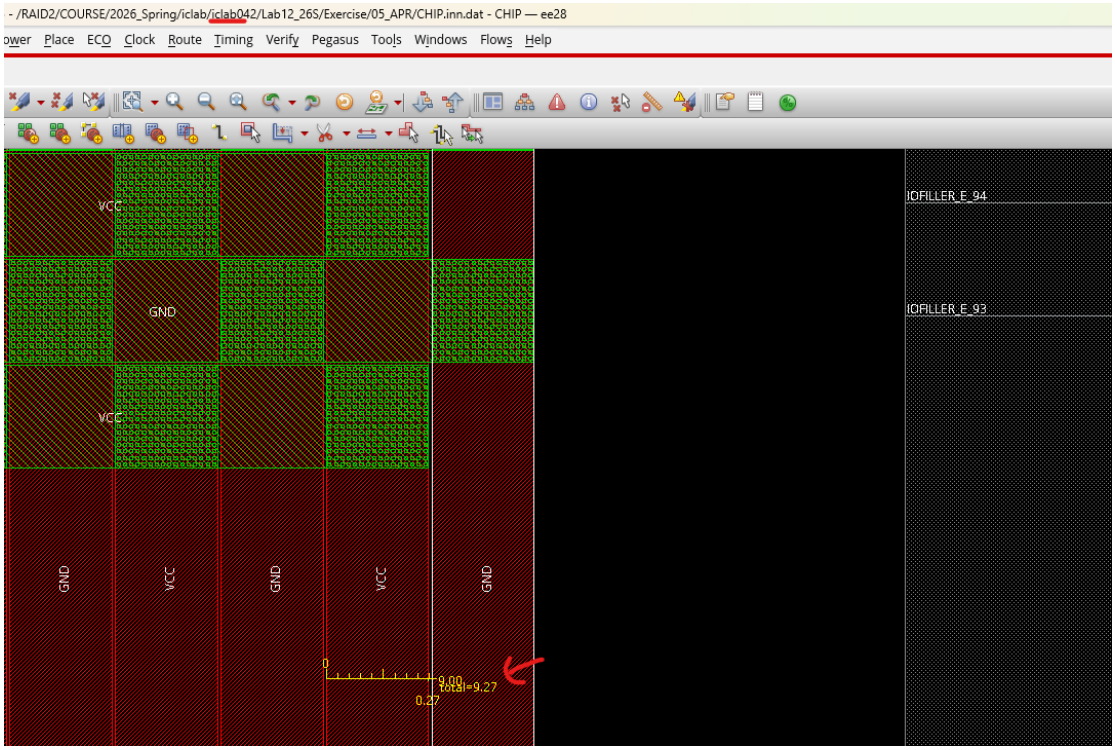
Horizontal core ring VCC



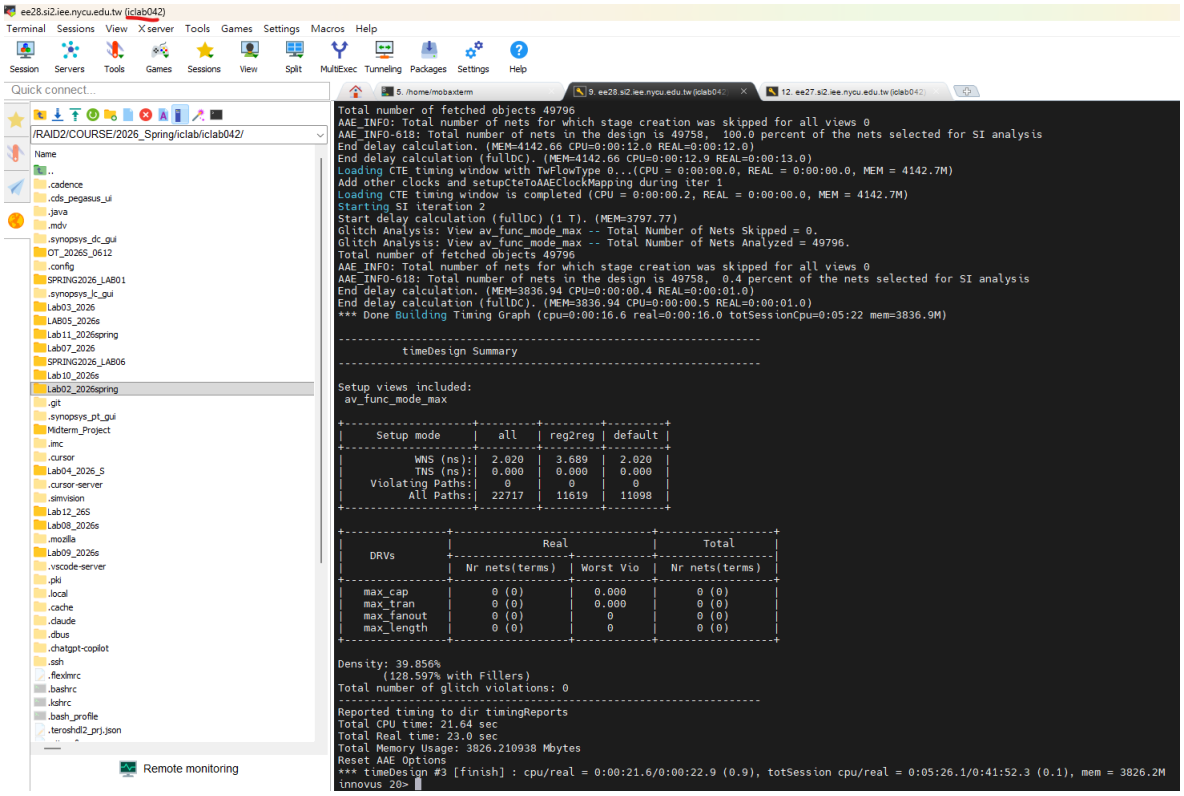
Vertical core ring GND



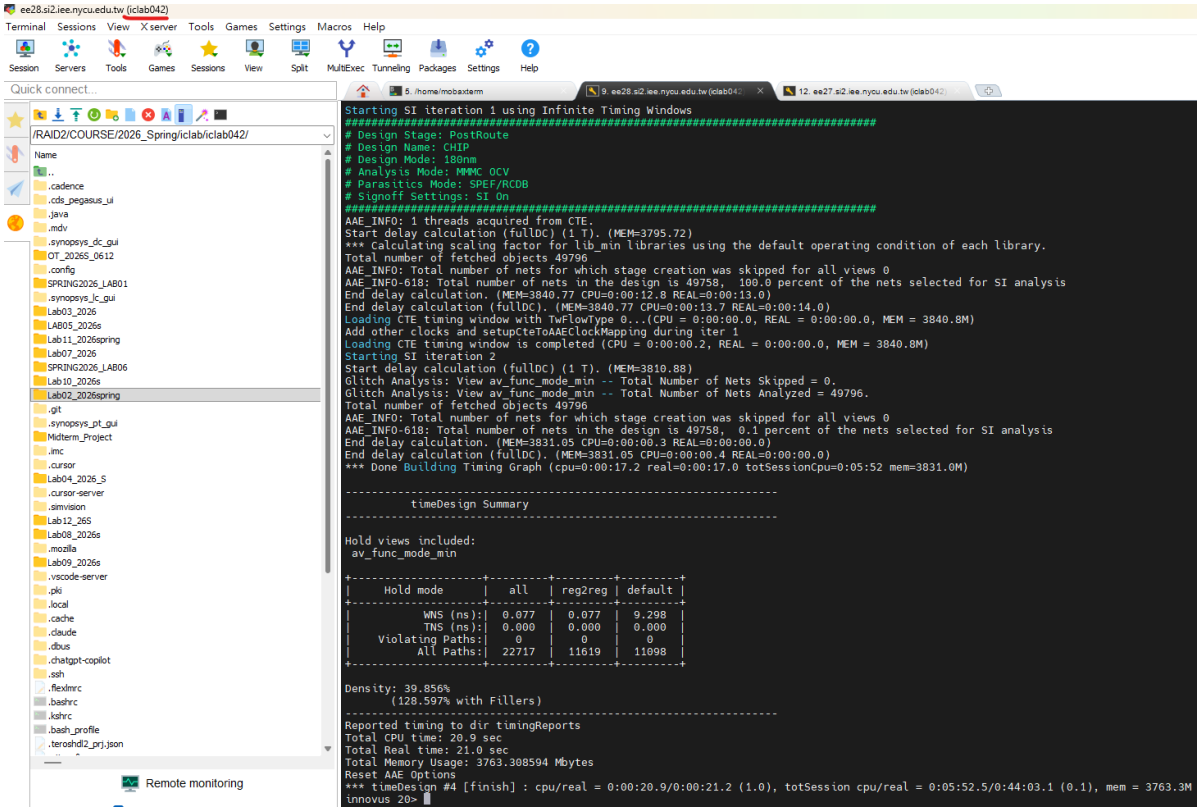
Vertical core ring VCC



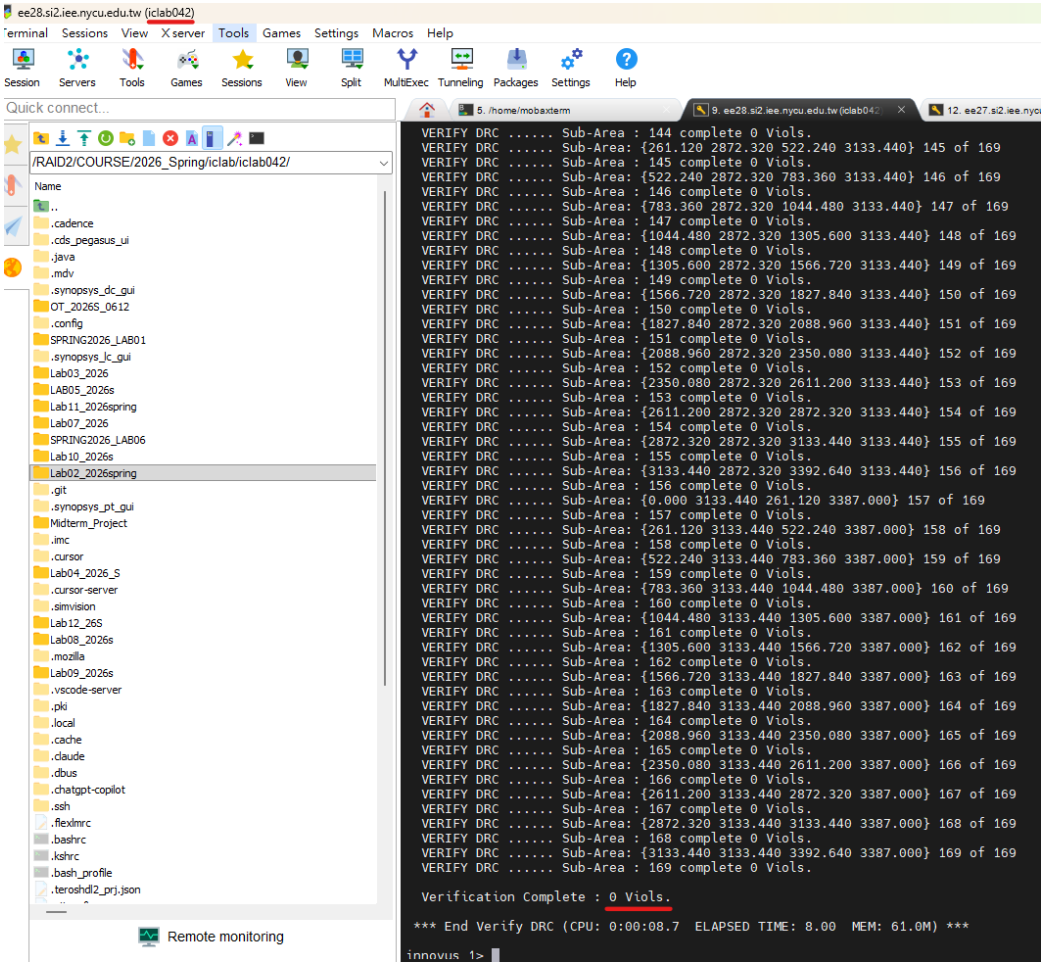
3. Post-Route setup time analysis :



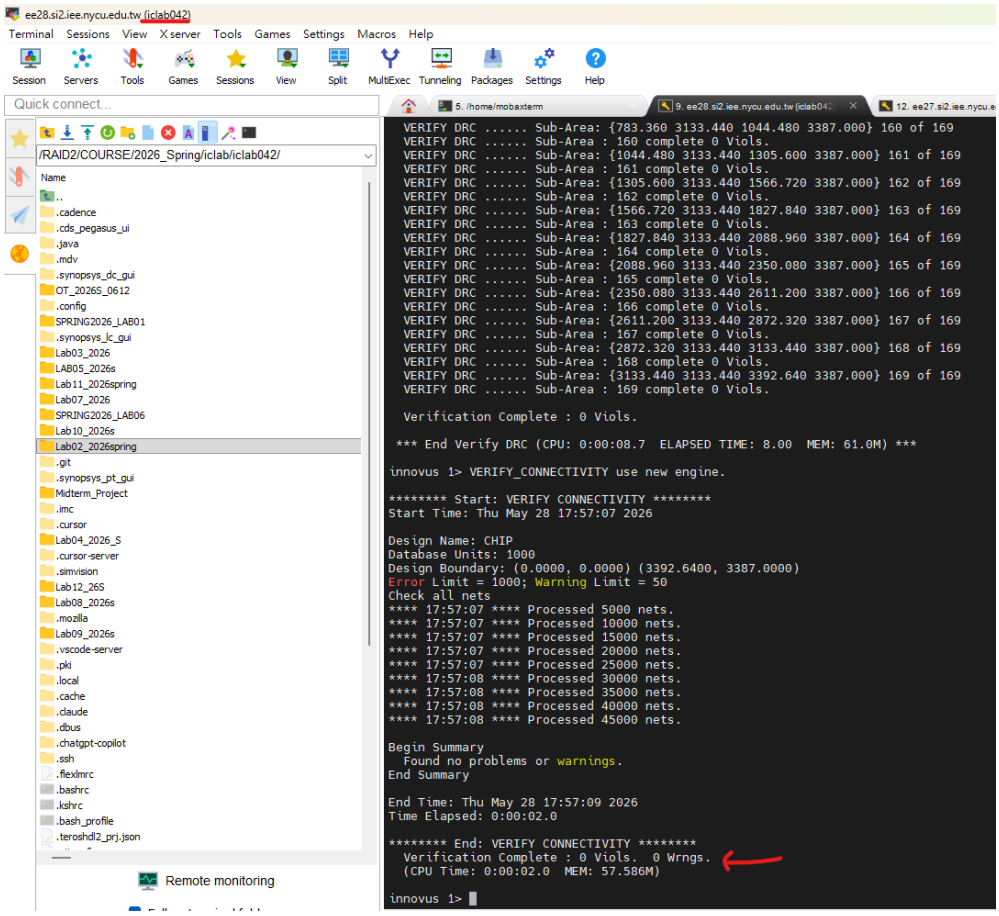
4. Post-Route hold time analysis :



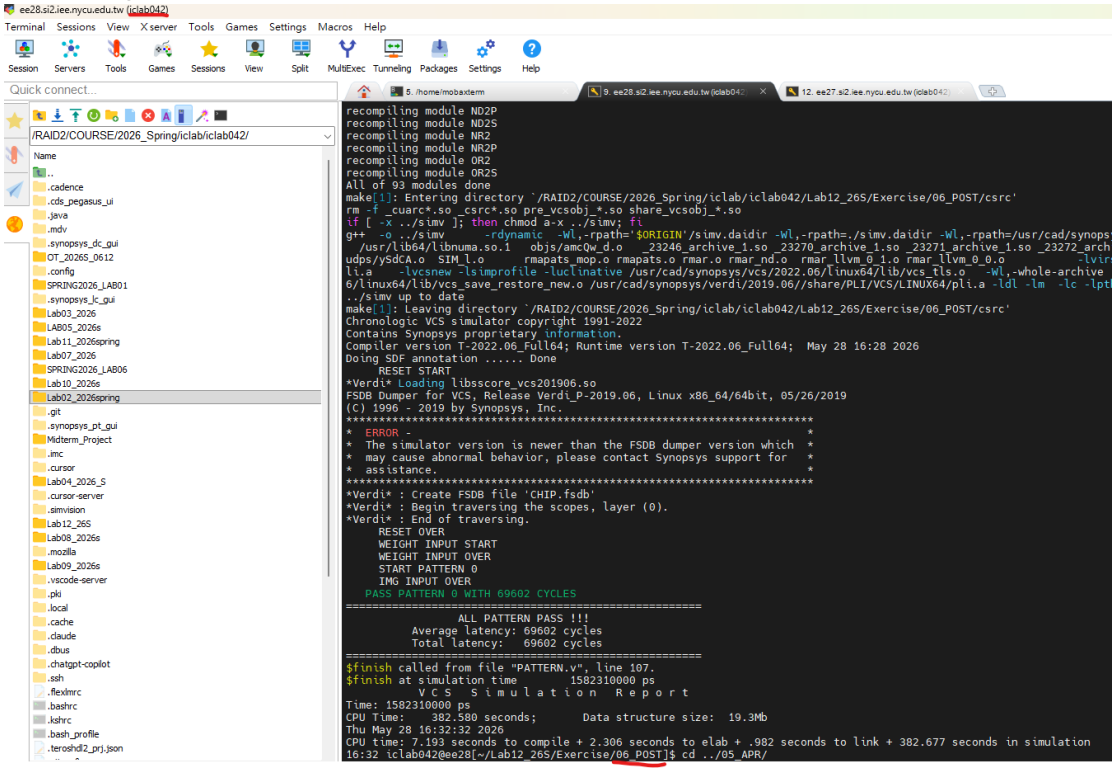
5. DRC result :


ee28.si2.iee.nycu.edu.tw (iclab042)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/RAID2/COURSE/2026_Spring/iclab/iclab042/
Name
..
.cadence
.cds_pegasus_ui
.java
.mdv
.synopsys_dc_gui
OT_2026S_0612
.config
SPRING2026_LAB01
.synopsys_lc_gui
Lab03_2026
LAB05_2026s
Lab11_2026spring
Lab07_2026
SPRING2026_LAB06
Lab10_2026s
Lab02_2026spring
.git
.synopsys_pt_gui
Midterm_Project
.imc
.cursor
Lab04_2026_S
.cursor-server
.simvision
Lab12_26S
Lab08_2026s
.mozilla
Lab09_2026s
.vscode-server
.pki
.local
.cache
.claude
.dbus
.chatgpt-copilot
.ssh
.flexlmrc
.bashrc
.kshrc
.bash_profile
.teroshdl2_prj.json
Remote monitoring
5. /home/mobaxterm
9. ee28.si2.iee.nycu.edu.tw (iclab042)
12. ee27.si2.iee.nycu
VERIFY DRC ..... Sub-Area : 144 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {261.120 2872.320 522.240 3133.440} 145 of 169
VERIFY DRC ..... Sub-Area : 145 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {522.240 2872.320 783.360 3133.440} 146 of 169
VERIFY DRC ..... Sub-Area : 146 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {783.360 2872.320 1044.480 3133.440} 147 of 169
VERIFY DRC ..... Sub-Area : 147 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1044.480 2872.320 1305.600 3133.440} 148 of 169
VERIFY DRC ..... Sub-Area : 148 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1305.600 2872.320 1566.720 3133.440} 149 of 169
VERIFY DRC ..... Sub-Area : 149 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1566.720 2872.320 1827.840 3133.440} 150 of 169
VERIFY DRC ..... Sub-Area : 150 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1827.840 2872.320 2088.960 3133.440} 151 of 169
VERIFY DRC ..... Sub-Area : 151 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2088.960 2872.320 2350.080 3133.440} 152 of 169
VERIFY DRC ..... Sub-Area : 152 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2350.080 2872.320 2611.200 3133.440} 153 of 169
VERIFY DRC ..... Sub-Area : 153 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2611.200 2872.320 2872.320 3133.440} 154 of 169
VERIFY DRC ..... Sub-Area : 154 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2872.320 2872.320 3133.440 3133.440} 155 of 169
VERIFY DRC ..... Sub-Area : 155 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3133.440 2872.320 3392.640 3133.440} 156 of 169
VERIFY DRC ..... Sub-Area : 156 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 3133.440 261.120 3387.000} 157 of 169
VERIFY DRC ..... Sub-Area : 157 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {261.120 3133.440 522.240 3387.000} 158 of 169
VERIFY DRC ..... Sub-Area : 158 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {522.240 3133.440 783.360 3387.000} 159 of 169
VERIFY DRC ..... Sub-Area : 159 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {783.360 3133.440 1044.480 3387.000} 160 of 169
VERIFY DRC ..... Sub-Area : 160 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1044.480 3133.440 1305.600 3387.000} 161 of 169
VERIFY DRC ..... Sub-Area : 161 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1305.600 3133.440 1566.720 3387.000} 162 of 169
VERIFY DRC ..... Sub-Area : 162 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1566.720 3133.440 1827.840 3387.000} 163 of 169
VERIFY DRC ..... Sub-Area : 163 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1827.840 3133.440 2088.960 3387.000} 164 of 169
VERIFY DRC ..... Sub-Area : 164 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2088.960 3133.440 2350.080 3387.000} 165 of 169
VERIFY DRC ..... Sub-Area : 165 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2350.080 3133.440 2611.200 3387.000} 166 of 169
VERIFY DRC ..... Sub-Area : 166 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2611.200 3133.440 2872.320 3387.000} 167 of 169
VERIFY DRC ..... Sub-Area : 167 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2872.320 3133.440 3133.440 3387.000} 168 of 169
VERIFY DRC ..... Sub-Area : 168 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3133.440 3133.440 3392.640 3387.000} 169 of 169
VERIFY DRC ..... Sub-Area : 169 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:08.7 ELAPSED TIME: 8.00 MEM: 61.0M) ***
innovus 1>

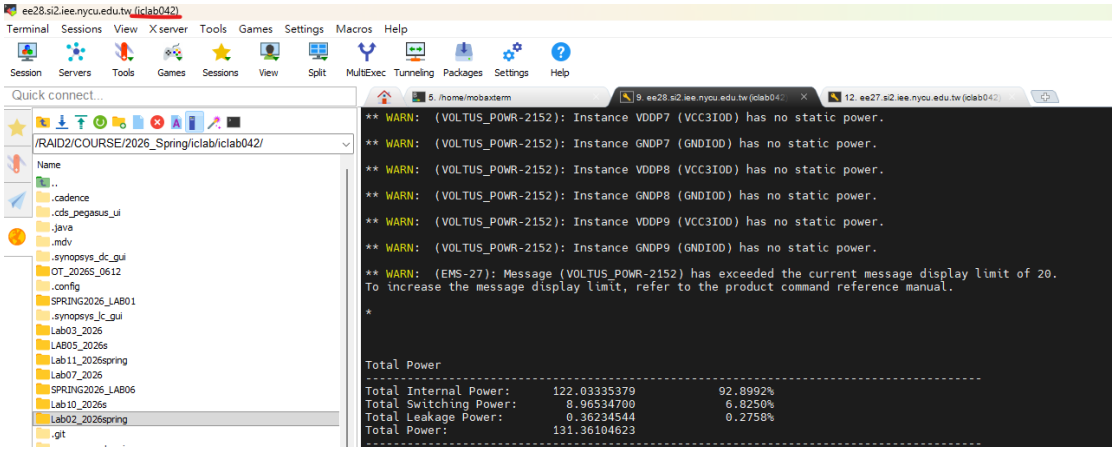
6. LVS result :



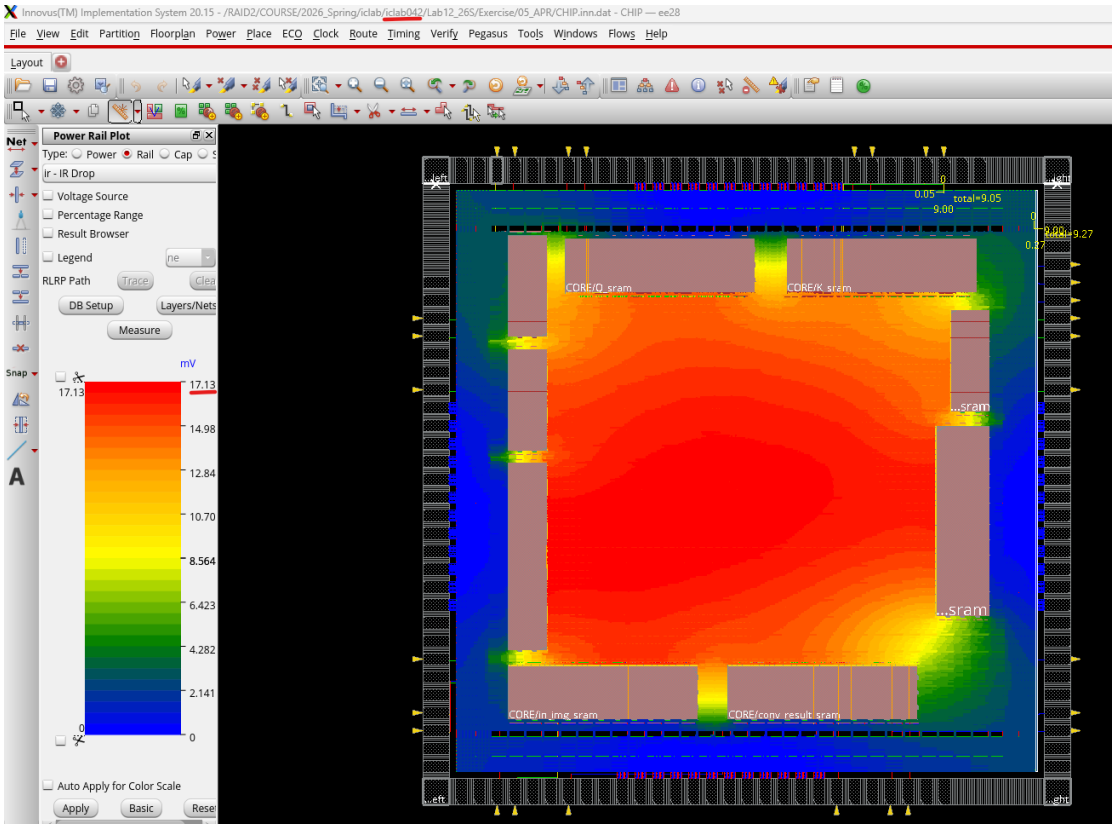
7. Post Layout simulation result :



8. Power result :



9. IR Drop Results :



IR Drop Solution:

First, I increased the number of core power pads to 24 pairs. Then, I used 10 pairs of core power rings to strengthen the power network around the core. In addition, I reduced the power stripe spacing to 90, which increased the number of stripes and allowed power to be distributed more evenly to the standard cells and SRAM regions.

For the floorplan, I adjusted the chip aspect ratio to be close to 1, making the

distance between the power pads and different circuit regions more balanced. This helped avoid larger voltage drops caused by overly long power delivery paths. After these optimizations, the final IR drop was successfully reduced to 17.13 mV, which is below the target limit of 50 mV.